18.06SC Unit 1 Exam

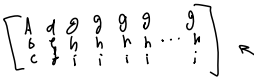
1 (24 pts.) This question is about an m by n matrix A for which

$$Ax = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ has } \textbf{no solutions} \quad \text{and} \quad Ax = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ has } \textbf{exactly one solution}.$$

(a) Give all possible information about m and n and the rank r of A.

(b) Find all solutions to $Ax = 0$ and **explain your answer**.

(c) Write down an example of a matrix A that fits the description in part (a).



a) the height of this matrix is 3, the width also is x since if more like the one above, then there still can be no solution for $\begin{vmatrix} 1 \\ 1 \\ 1 \end{vmatrix}$ and one for $\begin{vmatrix} 0 \\ 1 \\ 0 \end{vmatrix}$, but its safe to say that the width is 1s width, furthermore the r is $r < 3$ and $1 \leq r$ so the r is $1 \leq r < 3$.

b) only $x = 0$ works since the r does not equal the height of the matrix
or $x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

c) $A = \begin{vmatrix} 0 \\ 1 \\ 0 \end{vmatrix}$, $A = \begin{vmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 2 \end{vmatrix}$

2 (24 pts.) The 3 by 3 matrix A reduces to the identity matrix I by the following three row operations (in order):

E_{21}: Subtract 4 (row 1) from row 2.

E_{31}: Subtract 3 (row 1) from row 3.

E_{23}: Subtract row 3 from row 2.

(a) Write the inverse matrix A^{-1} in terms of the E's. **Then compute A^{-1}.**

(b) What is the original matrix A?

(c) What is the lower triangular factor L in $A = LU$?

$$A = \begin{vmatrix} A_1 & A_2 & A_3 \\ A_4 & A_5 & A_6 \\ A_7 & A_8 & A_9 \end{vmatrix} \qquad A = \begin{vmatrix} A_1 & A_2 & A_3 \\ A_4 - 4A_1 & A_5 - 4A_2 & A_6 - 4A_3 \\ A_7 - 3A_1 & A_8 - 3A_2 & A_9 - 3A_3 \end{vmatrix}$$

$$I = \begin{vmatrix} A_1 & A_2 & A_3 \\ A_4 - 4A_1 - A_7 + 3A_1 & A_5 - 4A_2 - A_8 + 3A_2 & A_6 - 4A_3 - A_9 + 3A_3 \\ A_7 - 3A_1 & A_8 - 3A_2 & A_9 - 3A_3 \end{vmatrix}$$

$A_1 = 1 \quad A_2 = 0 \quad A_3 = 0 \quad A_7 = 3 \quad A_4 = 4 \quad A_8 = 0 \quad A_5 = 1 \quad A_9 = 1$

$A_6 = 1$

$$A = \begin{vmatrix} 1 & 0 & 0 \\ 4 & 1 & 1 \\ 3 & 0 & 1 \end{vmatrix} \qquad \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 4 & 1 & 1 & 0 & 1 & 0 \\ 3 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -4 & 1 & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -4 & 1 & -1 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array}\right] \qquad A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & -1 \\ -3 & 0 & 1 \end{pmatrix}$$

3 (28 pts.) This 3 by 4 matrix depends on c:

$$A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 3 & c & 2 & 8 \\ 0 & 0 & 2 & 2 \end{bmatrix}$$

(a) *For each* c find a basis for the column space of A.

(b) *For each* c find a basis for the nullspace of A.

(c) *For each* c find the complete solution x to $Ax = \begin{bmatrix} 1 \\ c \\ 0 \end{bmatrix}$.

a) $\begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & c-3 & -4 & -4 \\ 0 & 0 & 2 & 2 \end{bmatrix}$ if $c=3$ then the column space is the first and the third columns of A $C(A) = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$ if $c \neq 3$ then the column space is the first, second and third columns $C(A) = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ c \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$

b) if $c \neq 3$ the free variable is the fourth column and setting $x_4 = 0$ gives us zero $Ax=0$ thus we set it to one and we get $N(A) = x \begin{bmatrix} -2 \\ 0 \\ 1 \\ 1 \end{bmatrix}$

if $c=3$ then the free variables are the 2nd and 4th columns so $N(A) = x_1 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix}$

c) for $c \neq 3$ it is $\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x \begin{bmatrix} -2 \\ 0 \\ 1 \\ 1 \end{bmatrix}$ for $c=3$ $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_1 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_1 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix}$

4 (24 pts.) (a) If A is a 3 by 5 matrix, what information do you have about the nullspace of A?

(b) Suppose row operations on A lead to this matrix $R = \texttt{rref}(A)$:

$$R = \begin{bmatrix} 1 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Write all known information about the columns of A.

(c) In the vector space M of all 3 by 3 matrices (you could call this a matrix space), what subspace S is spanned by all possible row reduced echelon forms R?

a) $2 \leq R(N(A)) \leq 5$

b) the first, fourth, and fifth columns are in R and are the column space

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